California Housing Price Predictor

Linear Regression Model — Task 1 Report

AI & Machine Learning Internship | Maincrafts Technology

1. Project Objective

This report documents the end-to-end machine learning pipeline built for Task 1 of the Maincrafts Technology AI & ML Internship. The goal is to train a **Linear Regression** model on the California Housing dataset to predict the **median house value** of a census district based on 8 socio-economic and geographic features. The full ML lifecycle is covered: data loading, EDA, preprocessing, training, evaluation, and reporting.

2. Dataset Overview

The California Housing dataset is derived from the 1990 US Census. It contains **20,640 records** with no missing values, making it clean and ready for modelling out-of-the-box.

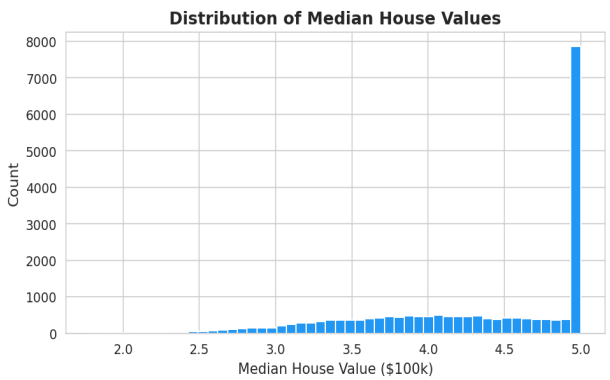
Feature	Type	Description
MedInc	Numeric	Median household income in block group ($10,000s)
HouseAge	Numeric	Median age of houses in the block group (years)
AveRooms	Numeric	Average number of rooms per household
AveBedrms	Numeric	Average number of bedrooms per household
Population	Numeric	Total population of the block group
AveOccup	Numeric	Average number of occupants per household
Latitude	Numeric	Latitude coordinate of the block group
Longitude	Numeric	Longitude coordinate of the block group
MedHouseVal	**TARGET**	**Median house value in $100,000s — what we predict**

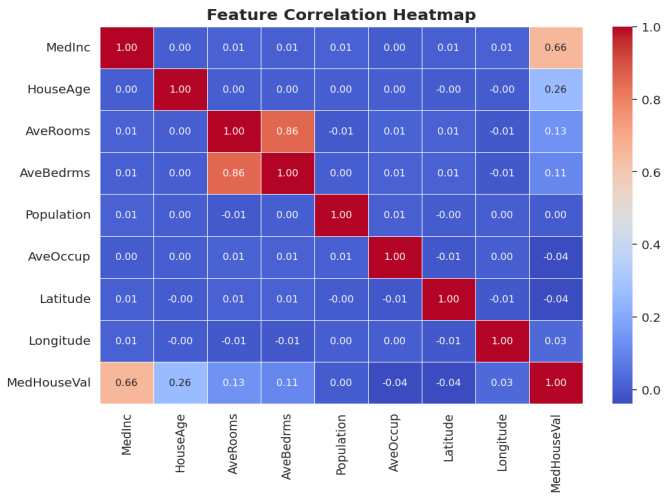
3. Exploratory Data Analysis (EDA)

EDA was performed to understand data distributions, feature relationships, and identify the most important predictors before training.

Key EDA Findings

- **No missing values** — all 20,640 records are complete; no imputation needed.
- **MedInc is the strongest predictor** with a correlation of **0.66** with MedHouseVal. Higher-income districts consistently have higher house values.
- **HouseAge** has a moderate positive correlation (0.26) — older California neighbourhoods tend to have higher property values.
- **Latitude** has a negative correlation (-0.04) — southern CA coastal areas command higher prices.
- **Population** and **AveOccup** have near-zero correlations and contribute little to prediction.
- The **target distribution** is left-skewed with a hard cap at $500,000 (5.0), a known artifact from the original census data collection.





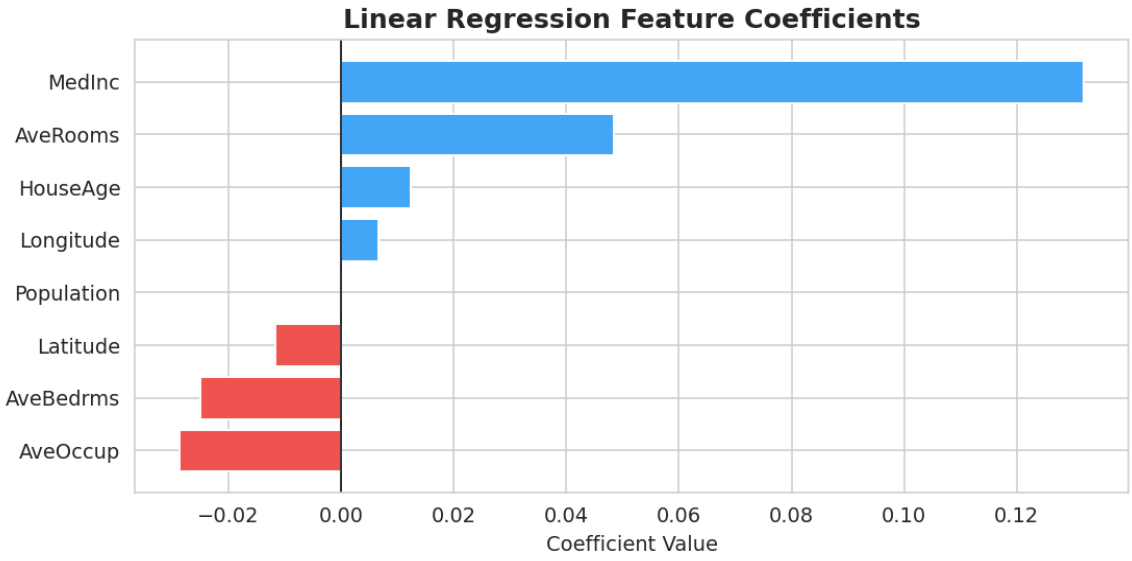
Left: Target distribution showing left-skew and $500k cap. Right: Correlation heatmap — MedInc dominates all features.

4. Model Training

A standard Ordinary Least Squares (OLS) **Linear Regression** was trained using scikit-learn. All 8 features were used without scaling (OLS is scale-invariant for predictions). An 80/20 stratified random split was applied: **16,512 training** and **4,128 testing** samples (random_state=42 for reproducibility).

Learned Coefficients

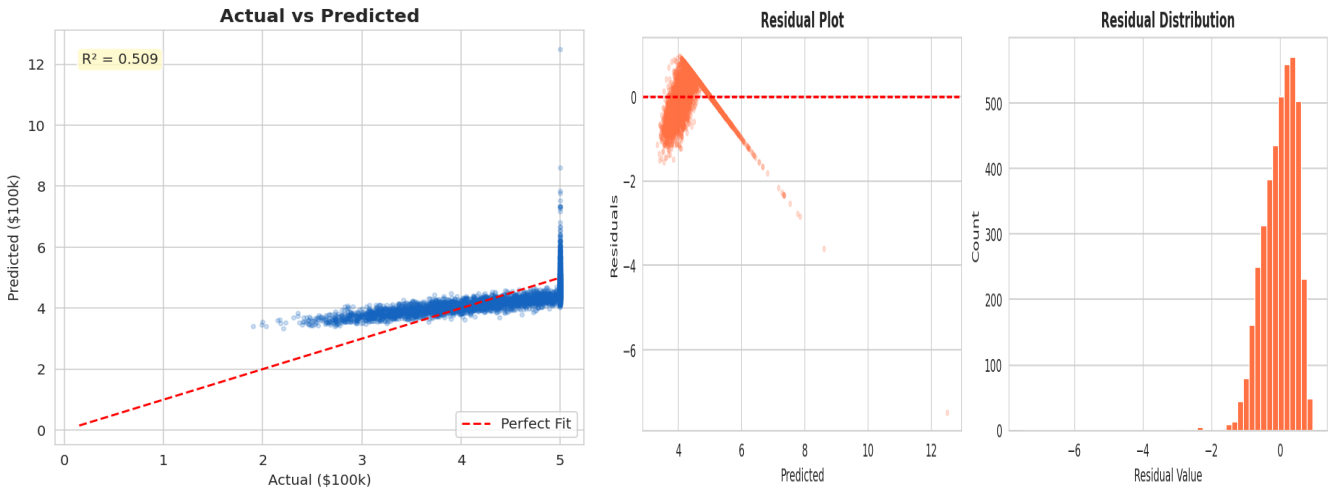
The chart below shows each feature's learned weight. **MedInc** has the largest positive coefficient, confirming income as the primary driver of house prices. Negative coefficients for AveBedrms and AveOccup suggest that denser occupancy correlates with lower-value areas.



5. Model Evaluation

The model was evaluated on the held-out test set using three standard regression metrics:

Metric	Formula	Value	Interpretation
MAE	mean|y - y_pred|	**0.3947**	Avg error ~$39,470 per district
RMSE	sqrt(mean(y-y_pred)^2)	**0.4992**	Penalises large errors more
R^2	1 - SS_res/SS_tot	**0.5087**	~51% of variance explained



Left: Actual vs Predicted scatter — points near the red line are accurate predictions. Right: Residuals are approximately normally distributed and centred at 0, confirming model validity.

6. Conclusions

Finding	Detail
Best Feature	MedInc (income) — correlation 0.66 with house value
Model Accuracy	$R^2 = 0.51$ — solid linear baseline, room to improve
Avg Error	$39,470 per district prediction (MAE)
Data Quality	Zero missing values — no imputation required
Residuals	Mean ≈ 0, roughly normal — assumptions partially met
Limitation	$500k cap in data introduces bias for high-value areas

7. Improvement Ideas

- **Ridge / Lasso Regression** — Add L2 or L1 regularisation to reduce overfitting and shrink irrelevant feature coefficients toward zero.
- **Feature Engineering** — Create new features like *BedroomRatio* (bedrooms/rooms), log-transform skewed features (Population, AveOccup), and polynomial terms for MedInc.
- **Outlier Treatment** — Cap extreme values in Population (max 35,682) and AveOccup (max 1,243) which distort the linear fit.
- **Non-linear Models** — Random Forest, Gradient Boosting (XGBoost / LightGBM) or Neural Networks typically achieve $R^2 > 0.82$ on this dataset.
- **Geographical Clustering** — Use KMeans to cluster Latitude/Longitude into regions (Bay Area, LA, Central Valley) and add as a categorical feature.
- **Cross-Validation** — Replace single 80/20 split with 5-fold or 10-fold CV for more reliable, variance-reduced performance estimates.
- **Feature Scaling** — Apply StandardScaler before training for better coefficient interpretability and compatibility with regularised models.

Model	Expected R^2	Complexity
Linear Regression	**~0.50**	**Low**
Ridge / Lasso	~0.52	Low
Linear + Poly Features	~0.65	Medium
Random Forest	~0.79	Medium
XGBoost / LightGBM	~0.84	Medium-High
Neural Network	~0.86+	High

**Current model (Linear Regression) highlighted in yellow. R^2 values are indicative.*